BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 103 · Characters and Images

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. Convert hexadecimal A3 to eight-bit binary.

Q2. How many different codes can 7 bits represent?
A. 7
B. 64
C. 128
D. 256

Q3. “Metadata is another name for the colour of every pixel.” Correct this. Explain why.

Q4. (a) A bitmap uses 5 bits per pixel. How many colours can it show? What if you add one bit? (b) A 50 × 20 pixel image doubles in width and height. Colour depth stays the same. How do pixel count and raw size change? Explain.

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I Do · Read new knowledge

Key vocabulary

Character set — Mapping between characters and numerical codes.

ASCII — A character encoding standard with a relatively limited basic character range.

Unicode — A universal standard covering many characters and writing systems.

Resolution — Width × height measured in pixels.

Colour depth — Bits used to represent each pixel's colour.

New learning

Characters need codes: Character sets map symbols to numerical codes stored in binary; ASCII and Unicode are examples.

Pixels build bitmaps: A bitmap stores the colour of each pixel in a grid. Resolution is width in pixels × height in pixels.

Colour depth: Colour depth is the number of bits per pixel. More bits allow more possible colours.

Image file size: Uncompressed size (bits) = width × height × colour depth; divide by eight for bytes (ignoring metadata).

Worked example

Calculate the uncompressed size of a 200 × 100 pixel bitmap with 8-bit colour depth. [4 marks]

1. Identify width and height

2. 200 × 100 = 20,000 pixels.

3. Multiply by colour depth: 20,000 × 8 = 160,000 bits.

4. Divide by eight: 20,000 bytes. [4]

We Do · Practise with guidance

Guided 1

A 100 × 80 image uses 16-bit colour. Calculate its uncompressed size in bytes. [4]

Worked method: 100×80=8000 pixels [1]; ×16=128,000 bits [1]; ÷8 [1] = 16,000 bytes [1].

Guided 2

Explain the impact of increasing colour depth from 8 to 16 bits. [4]

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Worked method: More possible colours per pixel [1] allows richer colour representation [1], but doubles uncompressed bits per pixel [1] and increases file size at equal resolution [1].

Guided 3

Explain one difference between ASCII and Unicode. [3]

Worked method: ASCII is a smaller character set [1]; Unicode supports a much wider range of characters and writing systems [1], improving international text support [1].

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You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Calculate the raw image data in bytes.

```
width = 10
height = 8
bitsPerPixel = 4
bytes = ...
print(bytes)
```

write

Question 1

An image is 50 pixels wide and 40 pixels high, using 8 bits per pixel. Calculate the uncompressed image size in bytes. [4] Source: Original OCR J277-style practice question.

Question 2

Explain why increasing an image's resolution increases file size when colour depth stays constant. [3] Source: Original OCR J277-style practice question.

Question 3

State one advantage of Unicode compared with basic ASCII. [2] Source: Original OCR J277-style practice question.

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Exit Ticket · Answer before checking

A trace table can help check how values change during a program.

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Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. 10100011: A is 1010 and 3 is 0011.

Q2. C — 128

Q3. Extra image details, such as width, height or colour depth. These are separate from pixel data.

Q4. (a) 32 colours; one extra bit doubles this to 64. (b) 1,000 pixels become 4,000. Raw size is four times larger: both dimensions double.

I Do / We Do pages · Guided practice

Q1. 100×80=8000 pixels [1]; ×16=128,000 bits [1]; ÷8 [1] = 16,000 bytes [1].

Q2. More possible colours per pixel [1] allows richer colour representation [1], but doubles uncompressed bits per pixel [1] and increases file size at equal resolution [1].

Q3. ASCII is a smaller character set [1]; Unicode supports a much wider range of characters and writing systems [1], improving international text support [1].

You Do pages · Code First

```
width = 10
height = 8
bitsPerPixel = 4
bytes = width * height * bitsPerPixel DIV 8
print(bytes)
```

Expected result: 40

You Do pages · Independent questions

Q1. 50×40=2000 pixels [1]; ×8=16,000 bits [1]; ÷8 [1] =2000 bytes [1].

Q2. More pixels are stored [1]; each pixel still requires the same number of bits [1]; total uncompressed bits increase [1].

Q3. Unicode represents a wider range of characters/scripts [1], supporting multilingual text [1].

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Exit Ticket · Model answer

True

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: